

Controlling, Information, and Technology

Target Skill

controlling skill: the ability to use information and technology to ensure that an event occurs as it was planned to occur

o b j e c t i v e s

To help build my *controlling skill*, when studying this chapter, I will attempt to acquire:

1. A definition of control
2. A thorough understanding of the controlling subsystem
3. Insights into the relationship between power and control
4. An understanding of the steps required to increase the quality of a controlling subsystem
5. An understanding of the relationship between data and information
6. Insights regarding the main factors that influence the value of information
7. An understanding of the importance of an information system (IS) to an organization
8. Insights regarding how to manage an IS effectively

CHALLENGE CASE

SPERRY VAN NESS: HARNESSING TECHNOLOGY FOR BUSINESS SUCCESS

COMPANIES THAT HANDLE INVENTORY often rely on information-gathering systems to help them in the controlling process. For commercial real estate brokerage firms, the “inventory” consists of real estate listings. Commercial real estate brokerage Sperry Van Ness uses the power of wireless technology to track inventory and maintain its competitive position.

Sperry Van Ness was founded in 1987 in Irvine, California, and since that time the organization has grown rapidly. Today, its nearly 1,000 real estate professionals work in the more than 150 Sperry Van Ness locations in the United States and overseas. The company offers a full range of real estate services, including brokerage, consulting, asset management, property management, leasing, accelerated marketing, and auction services.

The recent global economic downturn has had a chilling effect on the real estate market. For nearly two years, commercial lending in the United States has slowed almost to a stop, creating a weak market with more sellers anxious to unload properties than buyers interested in purchasing them. A construction boom that ended with the downturn resulted in a glut of office and commercial space, much of it now standing vacant while builders attempt to recoup their investments.

Sperry Van Ness has addressed the market challenges by using technology to better control its inventory, which will hopefully spur sales. It recently became one of the first organizations in the commercial real estate industry to adopt a mobile marketing platform called Qconnect.mobi. Working through the use of two-dimensional barcodes, Qconnect enables Sperry Van Ness to manage its commercial property listings. By scanning the barcodes with a smartphone, users can gain

immediate handheld access to Sperry Van Ness’s commercial listings around the world, including comprehensive information about the property, images, video, and more.

The Sperry Van Ness office in Chicago was the first to implement the new technology, using it to market 13,000 square feet of restaurant and retail space in a luxury high-rise development in Chicago’s trendy Fulton River district.

Here’s how the process works. As soon as a contract is signed, the Sperry Van Ness broker uses Qconnect to list the property. The listing becomes available instantaneously via mobile phone to brokers and investors, who can access all the information they might need about the property.

During a challenging economy, quick turnaround becomes an even more critical factor in real estate. For that reason, the Qconnect platform represents a competitive advantage for Sperry Van Ness. Said company president and CEO Kevin Maggiacomo, “This new mobile marketing platform is a result of our efforts to improve our efficiencies in marketing our listings.”¹



■ **Sperry Van Ness offers brokerage services for commercial real estate such as this retail and office space available for sale or lease.**

EXPLORING YOUR MANAGEMENT SKILL

You can explore your level of controlling skill **before** studying the chapter by completing the exercise “Exploring Your Management Skill: Part 1” on page 521 and **after** studying this

chapter by completing the exercise “Exploring Your Management Skill: Part 2” on page 522.

THE CONTROLLING CHALLENGE

According to the Challenge Case, issues at Sperry Van Ness involve maintaining and improving the speed of operations, internal communication, and communication with clients and prospects. The management function called *control* can help professionals at Sperry Van Ness and other organizations improve such issues, and the material in this chapter explains

why these activities would be considered controlling. The following material also elaborates on the control function as a whole. Major topics in this chapter are (1) fundamentals of controlling, (2) power and control, (3) information, and (4) information systems.

THE FUNDAMENTALS OF CONTROLLING

As the scale and complexity of modern organizations grow, so does the problem of control in organizations. Prospective managers, therefore, need a working knowledge of the essentials of the controlling function.² To this end, the following sections provide a definition of control, a definition of the process of controlling, and a discussion of the various types of control that can be used in organizations.

Defining Control

Stated simply, **control** entails ensuring that an event occurs as it was planned to occur.³ As implied by this definition, planning and control are virtually inseparable functions.⁴ In fact, these two functions have been called the Siamese twins of management. According to Robert L. Dewelt:

The importance of the planning process is quite obvious. Unless we have a soundly charted course of action, we will never quite know what actions are necessary to meet our objectives. We need a map to identify the timing and scope of all intended actions. This map is provided through the planning process.

But simply making a map is not enough. If we don't follow it or if we make a wrong turn along the way, chances are we will never achieve the desired results. A plan is only as good as our ability to make it happen. We must develop methods of measurement and control to signal when deviations from the plan are occurring so that corrective action can be taken.⁵

Murphy's Law is a lighthearted adage making the serious point that managers should continually control—that is, check to see that organizational activities and processes are going as planned. According to Murphy's Law, anything that can go wrong will go wrong.⁶ This law reminds managers to remain alert for possible problems, because even if a management system appears to be operating well, it might be eroding under the surface. Managers must always seek feedback on how the system is performing and make corrective changes whenever warranted.

Defining Controlling

Controlling is the process managers go through to control. According to Roberto Mockler, controlling is

a systematic effort by business management to compare performance to predetermined standards, plans, or objectives to determine whether performance is in line with these standards and presumably to take any remedial action required to see that human and other corporate resources are being used in the most effective and efficient way possible in achieving corporate objectives.⁷

Objectives

The controller (or comptroller) is responsible for all accounting activities within the organization.

Functions

1. *General accounting*—Maintain the company's accounting books, accounting records, and forms.
 - a. Preparing balance sheets, income statements, and other statements and reports
 - b. Giving the president interim reports on operations for the recent quarter and fiscal year to date
 - c. Supervising the preparation and filing of reports to the SEC
2. *Budgeting*—Prepare a budget outlining the company's future operations and cash requirements.
3. *Cost accounting*—Determine the cost to manufacture a product and prepare internal reports for management of the processing divisions.
 - a. Developing standard costs
 - b. Accumulating actual cost data
 - c. Preparing reports that compare standard costs to actual costs and highlight unfavorable differences
4. *Performance reporting*—Identify individuals in the organization who control activities and prepare reports to show how well or how poorly they perform.
5. *Data processing*—Assist in the analysis and design of a computer-based information system. Frequently, the data-processing department is under the controller, and the controller is involved in management of that department as well as other communications equipment.
6. *Other duties*—Other duties that may be assigned to the controller by the president or by corporate bylaws include:
 - a. Tax planning and reporting
 - b. Service departments such as mailing, telephone, janitors, and filing
 - c. Forecasting
 - d. Corporate social relations and obligations

FIGURE 21.1

Specific components of the controlling subsystem

For example, production workers generally have daily production goals. At the end of each working day, the number of units produced by each worker is recorded so weekly production levels can be determined. If these weekly totals are significantly below weekly goals, the supervisor must take corrective action to ensure actual production levels equal planned ones. If, on the other hand, production goals are being met, the supervisor should allow work to continue as it has in the past.⁸

The following sections discuss the controlling subsystem and provide more details about the control process itself.

The Controlling Subsystem As with the planning, organizing, and influencing functions described in earlier chapters, controlling can be viewed as a subsystem of the overall management system. The purpose of this subsystem is to help managers enhance the success of the overall management system through effective controlling. Figure 21.1 shows the specific components of the controlling subsystem.

The Controlling Process As Figure 21.2 illustrates, there are three main steps in the controlling process:

1. Measuring performance
2. Comparing measured performance to standards
3. Taking corrective action

MEASURING PERFORMANCE Before managers can determine what must be done to make an organization more effective and efficient, they must measure current organizational performance.⁹ However, before they can take such a measurement, they must establish some unit

the number of floors swept, the number of windows washed, or the number of light bulbs changed. After designating these units of measure, the manager has to determine the number of each of these units accomplished by each janitor. The process of determining both the units of measure and the number of units associated with each janitor furnishes the manager with a measure of janitorial performance.

- **What to Measure.** Managers must always keep in mind that a wide range of organizational activities can be measured as part of the control process. For example, the amounts and types of inventory on hand are commonly measured to control inventory, while the quality of goods and services being produced is commonly measured to control product quality. Performance measurements can relate as well to various effects of production, such as the degree to which a particular manufacturing process pollutes the atmosphere.

The degree of difficulty in measuring various types of organizational performance, of course, is determined primarily by the activity being measured. For example, it is far more difficult to measure the performance of a highway maintenance worker than to measure the performance of a student enrolled in a college-level management course.

COMPARING MEASURED PERFORMANCE TO STANDARDS Once managers have taken a measure of organizational performance, their next step in controlling is to compare this measure against some standard. A **standard** is the level of activity established to serve as a model for evaluating organizational performance.¹¹ The performance evaluated can be for the organization as a whole or for some individuals working within the organization.¹² In essence, standards are the yardsticks that determine whether organizational performance is adequate or inadequate.¹³

Studying operations at General Electric (GE) will give us some insights into the different kinds of standards managers can establish. GE has established the following standards:

1. **Profitability standards**—In general, these standards indicate how much money GE would like to make as profit over a given period—that is, its return on investment. More and more, GE is using computerized preventive maintenance on its equipment to help maintain profitability standards. Such maintenance programs have reduced labor costs and equipment downtime and thereby have helped raise company profits.
2. **Market position standards**—These standards indicate the share of total sales in a particular market that GE would like to have relative to its competitors. GE market position standards were set by company chairman John F. Welch, Jr., in 1988, when he announced that henceforth any product his company offers must achieve the highest or second-highest market share compared to similar products offered by competitors or it would be eliminated or sold to another firm.
3. **Productivity standards**—How much various segments of the organization should produce is the focus of these standards. Management at GE has found that one of the best ways to convince organization members to commit themselves to increasing company productivity is simply to treat them with dignity and make them feel they are part of the GE team.
4. **Product leadership standards**—GE intends to assume a leading position in product innovation in its field. Product leadership standards indicate what must be done to attain such a position. Reflecting this interest

how managers do it

Establishing Standards at General Electric

(continued)

in innovation, GE has pioneered the development of synthetic diamonds for industrial use. In fact, GE is considered the leader in this area, having recently discovered a method for making synthetic diamonds at a purity of 99.9 percent. In all probability, such diamonds will eventually be used as a component of super-high-speed computers.

5. **Personnel development standards**—Standards in this area indicate the type of training programs GE personnel should undergo to develop properly. GE's commitment to sophisticated training technology is an indication of the seriousness with which the company takes personnel development standards. Company training sessions are commonly supported by sophisticated technology such as large-screen projection systems, computer-generated visual aids, combined video and computer presentations, and laser videos.

(Continued)

6. **Employee attitudes standards**—These standards indicate what types of attitudes GE managers should strive to inculcate in GE employees. Like many other companies today, GE is trying to build positive attitudes toward product quality in its employees.
7. **Social responsibility standards**—GE recognizes its responsibility to make a contribution to society. Standards in this area outline the level and types of contributions management believes GE should make. One recent activity that reflects social responsibility standards at GE is the renovation of San Diego's Vincent de Paul Joan Kroc center for the homeless, accomplished by work teams made up of GE employees. These teams painted, cleaned, and remodeled a building to create a better facility for some of San Diego's disadvantaged citizens.
8. **Standards reflecting the relative balance between short- and long-range goals**—These standards express the relative emphasis that should be placed on attaining various short- and long-range goals. GE recognizes that short-range goals exist to enhance the probability that long-range goals will be attained. ■

Successful managers pinpoint all important areas of organizational performance and establish corresponding standards in each area.¹⁴ Managers at American Airlines, for example, have set two specific standards for appropriate performance of its airport ticket offices: (1) at least 95 percent of the flight arrival times posted should be accurate, meaning that actual arrival times do not deviate more than 15 minutes from posted times, and (2) at least 85 percent of customers coming to the airport ticket counter should not have to wait more than 5 minutes to be serviced.

TAKING CORRECTIVE ACTION After actual performance has been measured and compared with established performance standards, the next step in the controlling process is to take corrective action if necessary. **Corrective action** is managerial activity aimed at bringing organizational performance up to the level of performance standards.¹⁵ In other words, corrective action focuses on correcting organizational mistakes that are hindering organizational performance. Before taking any corrective action, however, managers should make sure the standards they are using were properly established and their measurements of organizational performance are valid and reliable.¹⁶

- **Recognizing Problems.** At first glance, it seems a fairly simple proposition that a manager should take corrective action to eliminate a **problem**—any factor within an organization that is a barrier to organizational goal attainment.¹⁷ In practice, however, it often proves

difficult to pinpoint the problem causing some undesirable organizational effect. Let us suppose that a performance measurement indicates a certain worker is not adequately passing on critical information to fellow workers. If the manager is satisfied that the communication standards are appropriate and that the performance measurement information is both valid and reliable, the manager should take corrective action to eliminate the problem causing this substandard performance.

- **Recognizing Symptoms.** What exactly is the problem causing substandard communication in this situation? Is it that the worker is not communicating adequately simply because he or she doesn't want to communicate? Is it that the job makes communication difficult? Is it that the worker does not have the necessary training to communicate in an appropriate manner? Before attempting to take corrective action, the manager must determine whether the worker's failure to communicate is a problem in itself or a **symptom**—a sign that a problem exists.¹⁸ For example, the worker's failure to communicate adequately could be a symptom of inappropriate job design or a cumbersome organizational structure.

Once the problem has been properly identified, corrective action can focus on one or more of the three primary management functions of planning, organizing, and influencing. That is, corrective action can include such activities as modifying past plans to make them more suitable for future organizational endeavors, making an existing organizational structure more suitable for existing plans and objectives, or restructuring an incentive program to ensure that high producers are rewarded more than low producers. Note that because planning, organizing, and influencing are closely related, it is likely that corrective action taken in one area will necessitate some corresponding action in one or both of the other two areas.

Many organizations use technology to help support their controlling activities. At discount retailer Stein Mart, for example, store managers use Oracle Retail, a suite of merchandising and planning tools, to help them minimize merchandising errors and take corrective action. Stein Mart management recently purchased and implemented the Oracle system to help the company carry out its goals to deliver fresh merchandise to every store and gain a thorough understanding of its unique customer base.¹⁹ ■

how managers do it

Using Technology
to Support Planning
at Stein Mart

POWER AND CONTROL

To control successfully, managers must understand not only the control process, but also how organization members relate to it. Up to this point, the chapter has emphasized the nonhuman variables of controlling. This section focuses on power, perhaps the most important human-related variable in the control process. The following sections present a definition of power, elaborate on the total power of managers, and list the steps managers can take to increase their power over other organization members.

A Definition of Power

Perhaps the two most often confused terms in management are *power* and *authority*. Authority was defined in Chapter 12 as the right to command or give orders. The extent to which an individual is able to influence others so they respond to orders is called **power**.²⁰ The greater this ability, the more power an individual is said to have.

Power and control are closely related. To illustrate, after comparing actual performance with planned performance and determining that corrective action is necessary, a manager usually gives orders to implement this action. Although the orders are issued by virtue of the manager's organizational authority, they may or may not be followed precisely, depending on how much power the manager has over the individuals to whom the orders are addressed.

Business and management scholar and Stanford University professor Jeffrey Pfeffer claims managers must learn how to wield power in order to advance their organization's agenda. In fact, he says, many highly competent professionals have floundered in management careers because they are uncomfortable using power.²¹ In the following sections, we describe the different types of power and describe steps managers may take to increase their power.

Total Power of a Manager

The **total power** a manager possesses is made up of two different kinds of power: position power and personal power. **Position power** is power derived from the organizational position a manager holds.²² In general, a manager moving from lower-level management to upper-level management accrues more position power. **Personal power** is power derived from a manager's relationships with others.²³

Steps for Increasing Total Power

Managers can increase their total power by enhancing either their position power or their personal power, or both. Position power is generally enhanced by a move to a higher organizational position, but most managers have little personal control over when they will move up in an organization. Managers do, however, have substantial control over the amount of personal power they hold over other organization members. John P. Kotter stresses the importance of developing personal power:

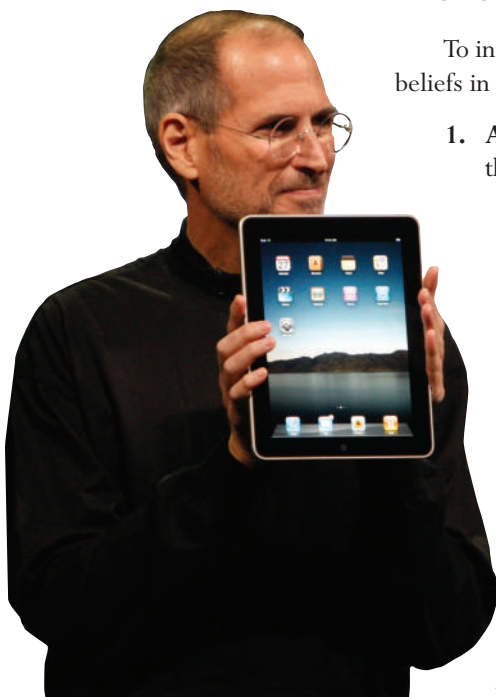
To be able to plan, organize, budget, staff, control, and evaluate, managers need some control over the many people on whom they are dependent. Trying to control others solely by directing them and on the basis of the power associated with one's position simply will not work—first, because managers are always dependent on some people over whom they have no formal authority, and second, because virtually no one in modern organizations will passively accept and completely obey a constant stream of orders from someone just because he or she is the “boss.”²⁴

To increase personal power, a manager should attempt to develop the following attitudes and beliefs in other organization members:²⁵

1. **A sense of obligation toward the manager**—If a manager succeeds in developing this sense of obligation, other organization members will allow the manager to influence them within certain limits. The basic strategy suggested for creating this sense of obligation is to do personal favors for people.
2. **A belief that the manager possesses a high level of expertise within the organization**—In general, a manager's personal power increases as organization members perceive that the manager's level of expertise is increasing. To raise perceptions of their expertise, managers must quietly make their significant achievements visible to others and build up a successful track record and a solid professional reputation.
3. **A sense of identification with the manager**—The manager can strive to develop this identification by behaving in ways that other organization members respect and by espousing goals, values, and ideals commonly held by them. The following description illustrates how a certain sales manager took steps to increase the degree to which his subordinates identified with him:

One vice president of sales in a moderate-sized manufacturing company was reputed to be so much in control of his sales force that he could get them to

Steve Jobs, Apple's CEO, gains a large part of his management power from his acknowledged technological expertise. Here he unveils the new iPad at the Yerba Buena Center for the Arts in San Francisco.



respond to new and different marketing programs in a third of the time taken by the company's best competitors. His power over his employees was based primarily on their strong identification with him and what he stood for. Immigrating to the United States at age seventeen, this person worked his way up "from nothing." When made a sales manager in 1965, he began recruiting other young immigrants and sons of immigrants from his former country. When made vice president of sales in 1970, he continued to do so. In 1975, 85 percent of his sales force was made up of people whom he hired directly or who were hired by others he brought.²⁶

4. **The perception that they are dependent on the manager**—The main strategy here is to clearly convey the amount of authority the manager has over organizational resources—not only those necessary for organization members to do their jobs, but also those organization members personally receive in such forms as salaries and bonuses. This strategy is aptly reflected in the managerial version of the Golden Rule: "He who has the gold makes the rules."

Making Controlling Successful

In addition to avoiding the potential barriers to successful controlling mentioned in the previous section, managers can perform certain activities to make the control process more effective. To increase the quality of the controlling subsystem, managers should make sure controlling activities take all the following factors into account.

Specific Organizational Activities Being Focused On Managers should make sure the various facets of the control process are appropriate to the control activity under consideration. For example, standards and measurements concerning a line worker's productivity are much different from standards and measurements concerning a vice president's productivity. Controlling ingredients related to the productivity of these individuals, therefore, must be different if the control process is to be applied successfully.

Different Kinds of Organizational Goals Control can be used for such different purposes as standardizing performance, protecting organizational assets from theft and waste, and standardizing product quality.²⁷ Managers should remember that the control process can be applied to many different facets of organizational life and that, if the organization is to receive maximum benefit from controlling, each of these facets must be emphasized.

Timely Corrective Action Some time will necessarily elapse as managers gather control-related information, develop necessary reports based on this information, and decide what corrective action should be taken to eliminate a problem. However, managers should take the corrective action as promptly as possible to ensure that the situation depicted by the information gathered has not changed. Unless corrective actions are timely, the organizational advantage of taking them may not materialize.

Communication of the Mechanics of the Control Process Managers should take steps to ensure people know exactly what information is required for a particular control process, how that information is to be gathered and used to compile various reports, what the purposes of the various reports actually are, and what corrective actions are appropriate given those reports. The lesson here is simple: For control to be successful, all individuals involved in controlling must have a working knowledge of how the control process operates.²⁸

ESSENTIALS OF INFORMATION

As mentioned in the previous sections, *controlling* is the process of making things happen as planned. Of course, managers cannot make things happen as planned if they lack information on the manner in which various events in the organization occur. The remainder of this chapter

L.L.Bean retail operations have grown from its original catalog business to include a Web site and a few retail stores like this one. Managers of each retail arm require different kinds of information. For instance, data about shoppers' access to technology and their privacy concerns are important to the company's Web site manager, while the demographics of the local labor supply matter to managers of the telephone centers and retail outlets.



discusses the fundamental principles of handling information in an organization by first presenting the essentials of information and then examining both information technology and information systems (IS).

The process of developing information begins with gathering some type of facts or statistics, called **data**. Once gathered, data typically are analyzed in some manner. In general terms, **information** refers to details about a project or the set of conclusions derived from data analysis. In management terms, however, information refers to the set of conclusions derived from the analysis of data that relate to the operation of an organization. Before a project commences, then, it is important for team members to reach agreement on what the term *information* will mean to them. In this way, they will be assured of operating from a mutual understanding.²⁹

As examples to illustrate the relationship between data and information, managers gather data regarding pay rates that individuals are receiving within industries to collect information about how to develop competitive pay rates, data regarding hazardous-materials accidents to gain information about how to improve worker safety, and data regarding customer demographics to gain information about product demand in the future.³⁰ Large organizations often use information systems to help them spot market trends and respond quickly and efficiently.³¹ In addition, many successful organizations are using the power of information systems to change the way they approach innovation.³²

The information that managers receive heavily influences managerial decision making, which, in turn, determines the activities that will be performed within the organization, which, in turn, dictates the eventual success or failure of the organization. Some management writers consider information to be of such fundamental importance to the management process that they define *management* as the process of converting information into action through decision making.³³ The next sections discuss the factors that influence the value of information and how to evaluate information.

Factors Influencing the Value of Information

Some information is more valuable than other information.³⁴ The value of information is defined in terms of the benefit that can accrue to the organization through its use.³⁵ The greater this benefit, the more valuable the information.

Four primary factors determine the value of information:

1. Information appropriateness
2. Information quality
3. Information timeliness
4. Information quantity

CHARACTERISTICS OF INFORMATION	OPERATIONAL CONTROL	MANAGEMENT CONTROL	STRATEGIC PLANNING
Source	Largely internal	————→	External
Scope	Well defined, narrow	————→	Very wide
Level of aggregation	Detailed	————→	Aggregate
Time horizon	Historical	————→	Future
Currency	Highly current	————→	Quite old/historical
Required accuracy	High	————→	Low
Frequency of use	Very frequent	————→	Infrequent

FIGURE 21.3

Characteristics of information appropriate for decisions related to operational control, management control, and strategic planning

In general, management should encourage generation, distribution, and use of organizational information that is appropriate, of high quality, timely, and of sufficient quantity. Following this guideline will not necessarily guarantee sound decisions, but it will ensure that important resources necessary to make such decisions are available.³⁶ Each of the factors that determines information value is discussed in more detail in the paragraphs that follow.

Information Appropriateness Information appropriateness is defined in terms of how relevant the information is to the decision-making situation the manager faces. If the information is quite relevant, then it is said to be appropriate. Generally, as the appropriateness of information increases, so does the value of that information.

Figure 21.3 shows the characteristics of information appropriate for the following common decision-making situations:³⁷

1. Operational control
2. Management control
3. Strategic planning

Operational Control, Management Control, and Strategic Planning Decisions *Operational control decisions* relate to ensuring that specific organizational tasks are carried out effectively and efficiently. *Management control decisions* relate to obtaining and effectively and efficiently using the organizational resources necessary to reach organizational objectives. *Strategic planning decisions* relate to determining organizational objectives and designating the corresponding action necessary to reach them.

As Figure 21.3 shows, characteristics of appropriate information change as managers shift from making operational control decisions to making management control decisions to making strategic planning decisions. Strategic planning decision makers need information that focuses on the relationship of the organization to its external environment, emphasizes the future, is wide in scope, and presents a broad view. Appropriate information for this type of decision is generally not completely current, but more historical in nature. In addition, this information does not need to be completely accurate because strategic decisions tend to be characterized by some subjectivity and focus on areas that are difficult to measure, such as customer satisfaction.

Information appropriate for making operational control decisions has dramatically different characteristics from information appropriate for making strategic planning decisions. Operational control decision makers need information that focuses, for the most part, on the internal organizational environment, emphasizes the performance history of the organization, and is well-defined, narrow in scope, and detailed. In addition, appropriate information for this type of decision is both highly current and highly accurate.

Information appropriate for making management control decisions generally has characteristics that fall somewhere between the extreme of appropriate operational control information and appropriate strategic planning information.

Information Quality The second primary factor that determines the value of information is **information quality**—the degree to which information represents reality. The more closely information represents reality, the higher the quality and the greater the value of that information. In general, the higher the quality of information available to managers, the better equipped managers are to make appropriate decisions and the greater the probability that the organization will be successful over the long term.

Perhaps the most significant factor in producing poor-quality information is *data contamination*. Inaccurate data gathering can result in information that is of low quality—a poor representation of reality.³⁸

Information Timeliness **Information timeliness**, the third primary factor that determines the value of information, is the extent to which the receipt of information allows decisions to be made and action to be taken so the organization can gain some benefit from possessing the information. Information received by managers at a point when it can be used to the organization's advantage is said to be timely.

For example, a product may be selling poorly because its established market price is significantly higher than the price of competitive products. If this information is received by management after the product has been discontinued, the information will be untimely. If, however, it is received soon enough to adjust the selling price of the product and thereby significantly increase sales, it will be timely.

Information Quantity The fourth and final determinant of the value of information is **information quantity**—the amount of decision-related information managers possess. Before making a decision, managers should assess the quantity of information they possess that relates to the decision being made. If this quantity is judged to be insufficient, more information should be gathered before the decision is made. If the amount of information is judged to be as complete as necessary, managers can feel justified in making the decision.

There is such a thing as *too much* information. According to Rick Feldcamp of Century Life of America, information overload—too much information to consider properly—can make managers afraid to make decisions and result in important decisions going unmade. Information overload is generally considered to be the major cause of indecision in organizations—commonly referred to as “paralysis by analysis.”³⁹

Evaluating Information

Evaluating information is the process of determining whether the acquisition of specified information is justified. As with all evaluations of this kind, the primary concern of management is to weigh the dollar value of benefit gained from using some quantity of information against the cost of generating that information.

Identifying and Evaluating Data According to the flowchart in Figure 21.4, the first major step in evaluating organizational information is to ascertain the value of that information by pinpointing the data to be analyzed, and then determine the expected value or return to be received from obtaining perfect information based on these data. Then this expected value is reduced by the amount of benefit that will not be realized because of deficiencies and inaccuracies expected to appear in the information.

Evaluating the Cost of Data Next, the expected value of organizational information is compared with the expected cost of obtaining that information. If the expected cost does not exceed the expected value, the information should be gathered. If it does exceed the

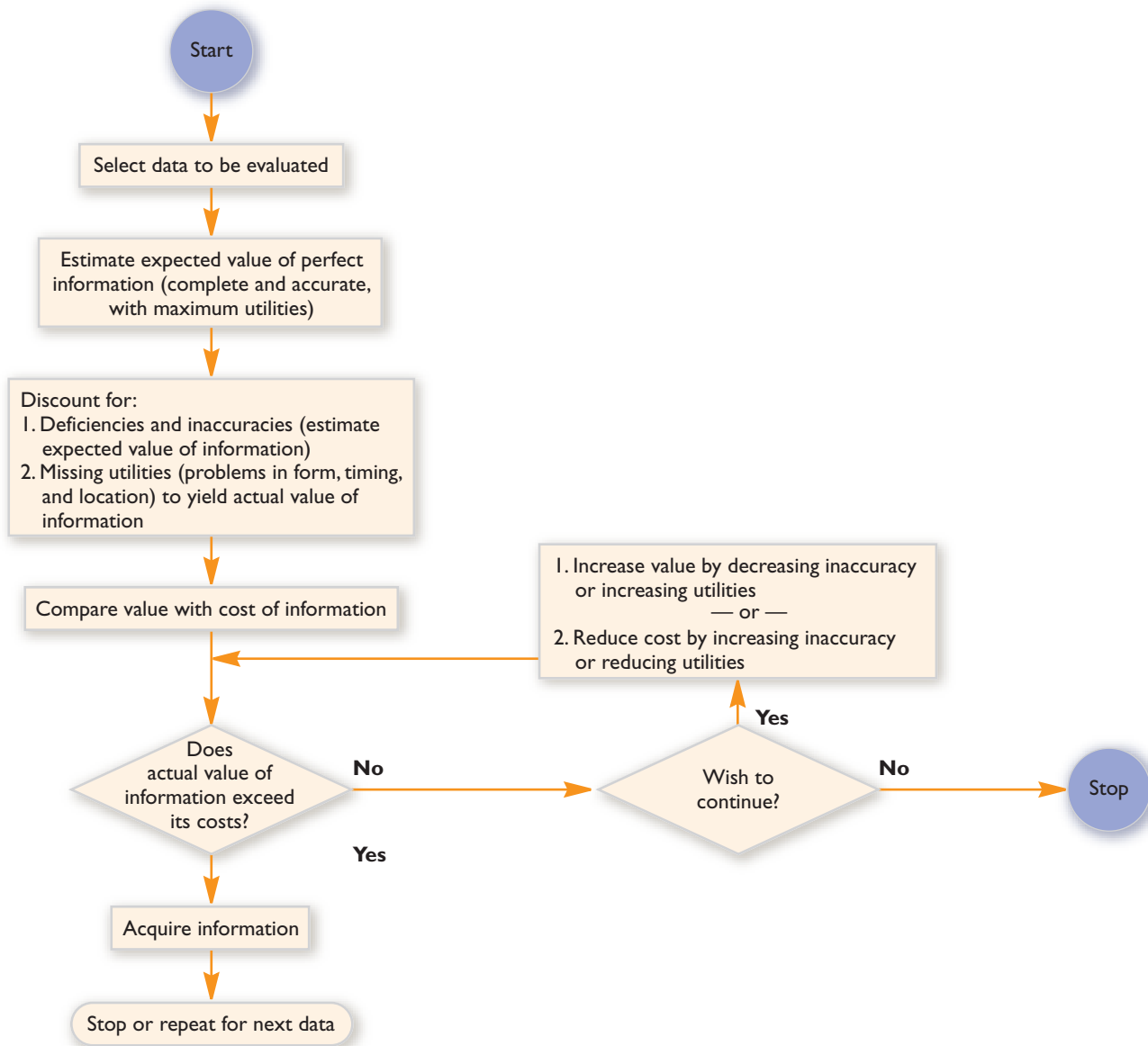


FIGURE 21.4 Flowchart of main activities in evaluating information

expected value, managers either must increase the information's expected value or decrease its expected cost before the information gathering can be justified. If neither of these objectives is possible, management cannot justify gathering the information.

INFORMATION TECHNOLOGY

Technology consists of any type of equipment or process that organization members use in the performance of their work. This definition includes tools as old as a blacksmith's anvil and tools as new and innovative as virtual reality. This section discusses one segment of technology, **information technology (IT)**, which is technology such as computers and telecommunication devices that focus on the use of information in the performance of work.

THE INFORMATION SYSTEM (IS)

In simple terms, an **information system (IS)** is a network of applications established within an organization to provide managers with information that will assist them in decision making.⁴⁰ The

following, more complete definition of an IS was developed by the Management Information System Committee of the Financial Executives Institute:

A system designed to provide selected decision-oriented information needed by management to plan, control, and evaluate the activities of the corporation. It is designed within a framework that emphasizes profit planning, performance planning, and control at all levels. It contemplates the ultimate integration of required business information subsystems, both financial and nonfinancial, within the company.⁴¹

The typical IS is a formally established organizational network that gives managers continual access to vital information. For example, the IS normally provides managers with ongoing reports relevant to significant organizational activities such as sales, worker productivity, and labor turnover. Based on information they gain via an IS, managers make decisions that are aimed at improving organizational performance. Because the typical IS is characterized by computer usage, managers can use an IS to gain online access to company records and condensed information in the form of summaries and reports. Overall, the IS is a planned, systematic mechanism for providing managers with relevant information in a systematic fashion.⁴²

The title of the specific organization member responsible for developing and maintaining an IS varies from organization to organization. In smaller organizations, a president or vice president may have this responsibility. In larger organizations, an individual with a title such as “director of information systems” or “chief information officer (CIO)” may be solely responsible for appropriately managing an entire IS department. The term *IS manager* is used in the sections that follow to indicate the person within the organization who has the primary responsibility for managing the IS. The term *IS personnel* is used to designate the nonmanagement individuals within the organization who possess the primary responsibility for actually operating the IS. Examples of nonmanagement individuals are computer operators and computer programmers. The sections that follow describe an IS more fully and outline the steps managers take to establish an IS.

Describing the IS

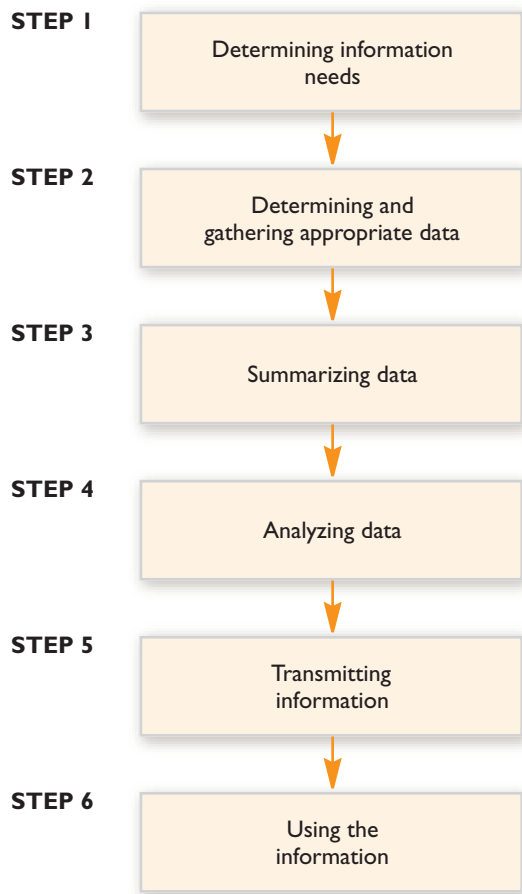
The IS is perhaps best described by a summary of the steps necessary to properly operate it,⁴³ and by a discussion of the different kinds of information various managers need to make job-related decisions.

Operating the IS IS personnel generally need to perform six sequential steps to properly operate an IS.⁴⁴ Figure 21.5 summarizes the steps and indicates the order in which they are performed. The first step is to determine what information is needed within the organization, when it will be needed, and in what form it will be needed. Because the basic purpose of the IS is to assist management in making decisions, one way to begin determining management information needs is to analyze the following:

1. Decision areas in which management makes decisions
2. Specific decisions within these decision areas that management must actually make
3. Alternatives that must be evaluated to make these specific decisions.

For example, insights regarding what information management needs in a particular organization can be gleaned by understanding that management makes decisions in the area of plant and equipment, that a specific decision related to this area involves acquiring new equipment, and that two alternatives that must be evaluated relating to this decision are buying newly developed, high-technology equipment versus buying more standard equipment that has been around for some time in the industry.

The second major step in operating the IS is pinpointing and collecting data that will yield needed organizational information. This step is just as important as determining information

**FIGURE 21.5**

The six steps necessary to operate an IS properly in order of their performance

needs of the organization. If collected data do not relate properly to information needs, it will be impossible to generate needed information.

After information needs of the organization have been determined and appropriate data have been pinpointed and gathered, summarizing the data and analyzing the data are, respectively, the third and fourth steps IS personnel generally should take to properly operate an IS. It is in the performance of these steps that IS personnel find computer assistance of great benefit.

The fifth and sixth steps are transmitting the information generated by data analysis to appropriate managers and getting the managers to actually use the information. The performance of these last two steps results in managerial decision making. Although each of the six steps is necessary if an IS is to run properly, the time spent on performing each step will naturally vary from organization to organization.

Different Managers Need Different Kinds of Information For maximum benefit, an IS must collect relevant data, transform that data into appropriate information, and transmit that information to the appropriate managers. Appropriate information for one manager within an organization, however, may not be appropriate information for another. Robert G. Murdick suggests that the degree of appropriateness of IS information for a manager depends on the activities for which the manager will use the information, the organizational objectives assigned to the manager, and the level of management at which the manager functions.⁴⁵ All of these factors are closely related. Murdick's thoughts on this matter are best summarized in Figure 21.6. As you can see from this figure, because the overall job situations of top managers, middle managers, and first-line managers are significantly different, the kinds of information these managers need to satisfactorily perform their jobs are also significantly different.

Organizational Level	Type of Management	Manager's Organizational Objectives	Appropriate Information from IS	How IS Information Is Used
1. Top management	CEO, president, vice president	Survival of the firm, profit growth, accumulation and efficient use of resources	Environmental data and trends, summary reports of operations, exception reports of problems, forecasts	Corporate objectives, policies, constraints, decisions on strategic plans, decisions on control of the total company
2. Middle management	Middle managers in such areas as marketing, production, and finance	Allocation of resources to assigned tasks, establishment of plans to meet operating objectives, control of operations	Summaries and exception reports of operating results, corporate objectives, policies, constraints, decisions on strategic plans, relevant actions and decisions of other middle managers	Operating plans and policies, exception reports, operating summaries, control procedures, decisions on resource allocations, actions and decisions related to other middle managers
3. First-line management	First-line managers whose work is closely related	Production of goods to meet marketing needs, supplying budgets, estimates of resource requirements, movement and storage of materials	Summary reports of transactions, detailed reports of problems, operating plans and policies, control procedures, actions and decisions of related first-line managers	Exception reports, progress reports, resource requests, dispatch orders, cross-functional reports

FIGURE 21.6 Appropriate IS information under various sets of organizational circumstances

Managing Information Systems

The effectiveness of an organization's IS depends largely on the ability of individuals within the organization to properly manage the IS. Three activities that improve IS effectiveness are managing user satisfaction, managing the IS workforce, and managing IS security. To the extent that employees are able to manage these activities, organizations will reap rewards in the form of IS effectiveness.

how managers do it

Scaling Data Systems for New Users at Sage

Just as different managers may require different kinds of information, not all information systems are appropriate for all organizations. For example, for-profit firms gauge their success in terms of profitability: do they make more money than they spend? And while nonprofits and government entities also track how their organizations use their funds, they evaluate their performance against different criteria, such as how well their stakeholders are served. To support the operations of nonprofits and government agencies, Sage North America developed its MIP Fund Accounting financial management software. Sage's recent introduction of HR modules to complement the software system enables users to automate their human resources activities with employee self-service, payroll, and tax compliance capabilities.⁴⁶ ■

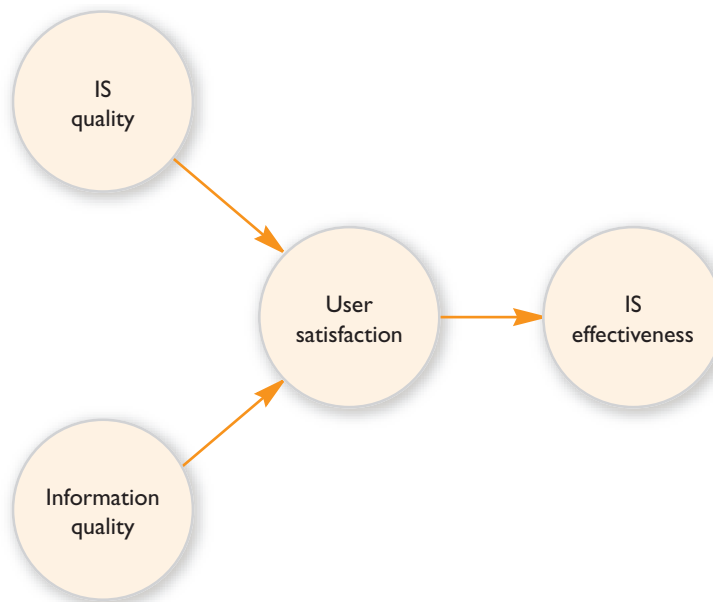


FIGURE 21.7
A model of IS effectiveness

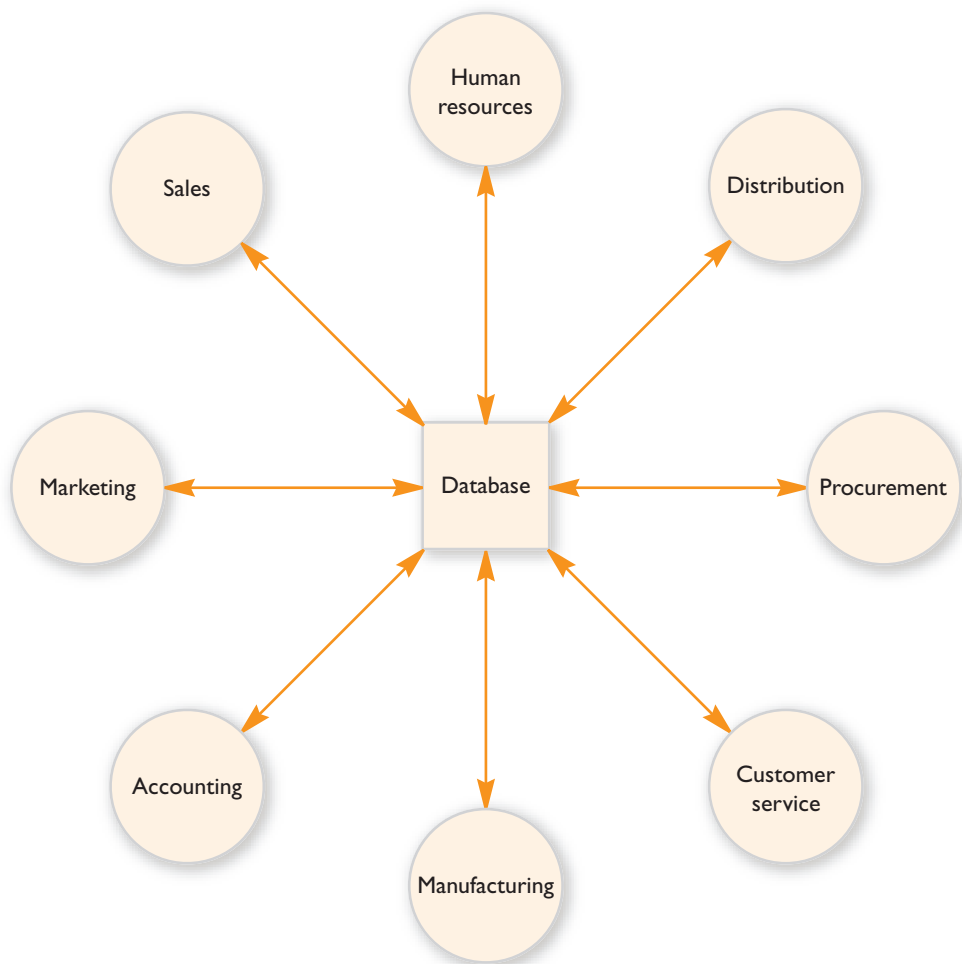
Managing User Satisfaction⁴⁷ One of the most important determinants of IS effectiveness is the degree to which employees, or users, are satisfied with the IS. User satisfaction, which denotes the degree of user satisfaction with the IS, is determined by two main factors: (1) the quality of the IS and (2) information quality. The quality of the IS refers to its ease of use. If a company's employees consider an IS easy to use, that IS would be labeled a high-quality IS. Information quality, on the other hand, measures the degree to which the information produced by the IS is accurate and in a format required by the user. Taken together, then, users are satisfied with the IS when the IS is of high quality and provides high-quality information.

User satisfaction is important because of its direct influence on IS effectiveness. When users are satisfied with the IS, the IS will be more integrated in the user's work routine, and the user will become more dependent on the IS. Stated differently, when employees are satisfied with the IS, they will use the IS in their work routines. As users become dependent on the IS and integrate the IS into their routines, the IS becomes effective. These relationships are depicted in Figure 21.7.

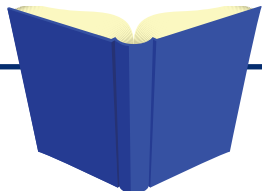
Managing the IS Workforce In recent years, executives have faced different obstacles in managing the IS workforce. During the economic and technological boom of the late 1990s, executives faced tremendous hurdles in terms of hiring and retaining IS employees.⁴⁸ During that period, as compared to other professionals, IS professionals were considered more difficult to hire and retain because the economic and technological boom created a multitude of job prospects for IS professionals. Moreover, executives found it quite expensive to replace IS professionals who left; some estimates suggest that the cost of replacing an IS employee is 1 to 2.5 times their annual salary.⁴⁹

In more recent years, however, companies have started to use workers in other countries to staff IS departments—and many expect this trend to continue. Although approximately 27,000 IS jobs in the United States were outsourced to other countries in 2000, experts expect this number to increase to almost 500,000 by 2015.⁵⁰ A recent survey indicated that almost half of firms outsource work to workers in other countries to reap cost advantages.⁵¹ Specifically, the cost of IS employees in international countries is much less than for IS employees in the United States (see Figure 21.8). This issue of lower costs partly explains why EDS, a Texas-based firm that offers its clients IT-based solutions, employs about 1,000 IS workers in India but expects this number to increase to nearly 20,000 in the near future.

Despite the cost advantages associated with these international workers, this practice creates other problems such as integrating domestic and nondomestic workforces, managing international languages and cultures, and defining global work expectations. In addition, U.S. firms could face a backlash from their customers for outsourcing IS work to other countries, as some customers could view the practice as being unpatriotic.

**FIGURE 21.8**

Centralized databases:
An illustration



class discussion highlight

MODERN RESEARCH AND CONTROLLING SKILL

Managing IS Road Warriors

Organizations must find ways to stay current with the most recent technological advances. Due to this need, the IS consulting industry has continued to develop in recent years. These consulting companies will send to a given organization an individual or team of consultants to help the organization implement the most advanced information systems. These consultants will often travel to new cities and stay for days, weeks, or months at a time.

Although this practice of employing IS consultants continues to grow, little is known about the consultants who travel to work with other organizations. To better understand these individuals, Professor Ahuja and colleagues examined a sample of 171 "road warriors." According to their framework,

road warriors are consultants who spend most of the week in a different city working with another organization and then return home for the weekend.

The researchers were particularly interested in the factors that influence the road warriors' commitment to their home organizations as well as their general levels of exhaustion. As you think about this unique set of individuals, which factors might influence their overall level of work exhaustion? Which factors might influence their commitment to their home organization (i.e., the consulting firms)? Why?

Source: This discussion is based on M. K. Ahuja, K. M. Chudoba, C. J. Kacmar, D. H. McKnight, and J. F. George, "IT Road Warriors: Balancing Work-Family Conflict, Job Autonomy, and Work Overload to Mitigate Turnover Intentions," *MIS Quarterly* 31, no. 1 (2007): 1–17.

TABLE 21.1 GASSP's Key Principles for Maintaining Information Security

Accountability principle	Organizations must clearly define and acknowledge information security accountability and responsibility.
Ethics principle	Organizations should use information and execute information security in an ethical manner.
Timeliness principle	Organizations should act in a timely manner to prevent or respond to breaches of and threats to information systems.
Assessment principle	Organizations should periodically assess the risks to information and information systems.
Equity principle	Management shall respect the rights of all employees when setting policy regarding security measures.

Managing IS Security As corporations rely more heavily on information systems, they become more susceptible to security issues involving these systems. In particular, companies may lose valuable financial, employee, or customer data due to security breaches involving IS. In addition, companies become increasingly vulnerable to viruses, worms, and trojan horses designed to paralyze information systems. As technology continues to change rapidly, it becomes more difficult for IS employees to prevent and eliminate these security threats.

In response to the increasing threat of security issues to information systems, private and public organizations around the world came together in 1992 to form the International Information Security Foundation. This committee produced a document known as the Generally Accepted System Security Principles (GASSP), which included a set of best practices for IS managers.⁵² The best practices listed within the GASSP provide a good starting point for managers when they are attempting to prevent security threats. Table 21.1 provides an overview of some of the broad principles outlined in the GASSP.

CHALLENGE CASE SUMMARY

The information in this chapter supports the notion that the efficient use of technology actually should be categorized as a control problem. Control is making things happen at Sperry Van Ness in a way they were planned to happen. Going one step further, the process of controlling is the action that management takes to control. Ideally, this process at Sperry Van Ness, as within any company, would include a determination of company plans, standards, and objectives so steps can be taken to eliminate company characteristics that caused deviation from these factors.

In theory, Sperry Van Ness management should view controlling activities within the company as a subsystem of the organization's overall management system. For management to achieve organizational control, the controlling subsystem requires a portion of the people, money, and other resources available within the company.

The process portion of the controlling subsystem at Sperry Van Ness involves three steps:

1. Measuring the performance levels of various selling units
2. Comparing these performance levels to predetermined performance standards for these units
3. Taking any corrective action necessary to make sure that planned performance levels are consistent with actual performance levels

Based on information in the Challenge Case, one area in which management should emphasize standards is in the arena of desired profitability. Management is initiating a new set of processes to ensure Sperry Van Ness and its real estate advisors in the field are working with the same information.

As the company gathers more information and considers potential corrective actions, management must be certain the action is aimed at organizational problems rather than at symptoms of problems. For Sperry Van Ness's management to be successful in controlling, they have to be aware not only of the intricacies of the control process itself, but also of how to deal with people as they relate to the control process. With regard to people and control, managers must consider the amount of power they hold over organization members.

The total amount of power that Sperry Van Ness management possesses comes from the positions they hold and from their personal relationships with other organization members. For example, the top managers already have

more position power than any other managers in the organization. Therefore, to increase their total power, they would have to develop their personal power. Top management might attempt to expand personal power by developing:

1. A sense of obligation in other organization members toward top managers
2. The belief in other organization members that top management has a high level of task-related expertise
3. A sense of identification that other organization members have with top management
4. The perception in organization members that they are dependent on top management

Information at Sperry Van Ness can be defined as conclusions derived from the analysis of data relating to the way in which the company operates. The case implies that managers at Sperry Van Ness will be better able to make sound decisions, including better control decisions, because of the successful data handling achieved by its information system. One important factor in evaluating the overall worth of Sperry Van Ness's information handling system would be the overall impact of the system on the value of information that company managers would receive. A manager such as Kevin Maggiacomo must see that investing in technology at a reasonable cost can enhance the value of information he receives and improve the appropriateness of decisions. That is, investments in improving information system components can enhance the appropriateness, quality, timeliness, and quantity of information that Maggiacomo can use to make decisions. Maggiacomo must believe and act on the notion that the benefits of making investments in technology will significantly improve his decisions.

For a company such as Sperry Van Ness to get maximum benefit from computer assistance, management must appropriately build each main ingredient of its IS. The IS the organizational network established to provide managers with information that helps them make job-related decisions. Such a system would normally necessitate the use of several IS personnel who would help determine information needs at the company, help determine and collect appropriate Sperry Van Ness data, summarize and analyze these data, transmit analyzed data to appropriate

Sperry Van Ness managers, and generally help managers in interpreting received IS information.

To make sure managers get appropriate information, Sperry Van Ness's IS personnel must appreciate how different managers need different kinds of information. As an example, a top manager would normally need information that summarizes trends in consumer tastes, competitor moves, and summary reports for productivity and costs related to various organizational units. Middle managers would need information that focuses more on specific operating units within the company, such as all specifics regarding the performance of a specific office. More lower-level managers would normally need information about sales goals.

Assume that Maggiacomo has just decided to establish a new IS. Sperry Van Ness, like any other company, would probably gain significantly by carefully planning the way in which its IS would be established. For example, perhaps the answers to the following questions, such as those that arise during the planning stage of Sperry Van Ness's IS, would be useful: Is an appropriate computer-based system being acquired and integrated? Does the company need new IS personnel, or will present personnel require further training to operate the new IS? Will managers need additional training to operate the new IS?

About the design and implementation stages of Sperry Van Ness's new IS, Maggiacomo should seek answers to such questions as: How do we design the new IS based on managerial decision making? How can we ensure the new IS as designed and implemented will actually exist and be functional?

Maggiacomo as well as IS personnel should continually try to improve the new IS. All users of the IS should be aware of the symptoms of an inadequate IS and should constantly attempt to pinpoint and eliminate corresponding weaknesses. Suggestions for improving the new IS could include (1) building additional cooperation between IS managers, IS personnel, and line managers; (2) stressing that the purpose of the IS is to provide managers with decision-related information; (3) using cost-benefit analysis to evaluate IS activities; and (4) ensuring that the IS operates in a people-conscious manner.

MANAGEMENT SKILL ACTIVITIES

This section is specially designed to help you develop controlling skill. An individual's controlling skill is based on an understanding of controlling concepts and the ability to apply those concepts in management situations. As a result, the following activities are designed both to heighten your understanding of controlling concepts and to help you gain facility in applying these concepts in various management situations.

UNDERSTANDING CONTROLLING CONCEPTS



To check your understanding and to practice using the concepts in this chapter, go to www.mymanagementlab.com and explore the material associated with Chapter 21.

Know Key Terms

Understanding the following key terms is critical to your understanding of chapter material. Define each of these terms. Refer to the page(s) referenced after a term to check your definition or to gain further insight regarding the term.

control 502	total power 508	information timeliness 512
controlling 502	position power 508	information quantity 512
standard 505	personal power 508	technology 513
corrective action 506	data 510	information technology (IT) 513
problem 506	information 510	information system (IS) 513
symptom 507	information appropriateness 511	
power 507	information quality 512	

Know How Management Concepts Relate

This section is comprised of activities that will further sharpen your understanding of management concepts. Answer essay questions as completely as possible. Also, remember that many additional true/false and multiple choice questions appear online at MyManagementLab.com to help you further refine your understanding of management concepts.

1. Describe the three main steps in the controlling process.
2. Define power and describe the determinants of an individual's power within an organization.
3. What is the difference between a problem and a symptom? Use an example to support your response.
4. Describe the six steps involved with information system performance.
5. What are the determinants of information system effectiveness?

DEVELOPING MANAGEMENT SKILLS

Learning activities in this section are aimed at helping you develop your controlling skill. Learning activities include Exploring Your Management Skill: Parts 1 and 2, Experiential Exercises, Cases, and a VideoNet Exercise.

Exploring Your Management Skill: Part 1

Before studying this chapter, respond to the following questions regarding the type of advice you would give to Sperry Van Ness CEO Kevin Maggiacomo, referenced in the Challenge Case. Then address the concerning controlling challenges that he presently faces within the company. You are not expected to be a controlling expert at this point. Answering the questions now can help you focus on important points when you study the chapter. Also, answering the questions again after you study the chapter will give you an idea of how much you have learned.

Record your answers here or go to MyManagementLab.com. Recording your answers in MyManagementLab will allow you to get immediate results and see how your score compares to your classmates. If you answer the questions in the book, the answers are located in the Exploring Your Management Skill Appendix at the end of the book.

FOR EACH STATEMENT CIRCLE:

- "Y" if you would give the advice to Maggiacomo.
- "N" if you would NOT give the advice to Maggiacomo.
- "NI" if you have no idea whether you would give the advice to Maggiacomo.

Mr. Maggiacomo, in meeting your controlling challenges at Sperry Van Ness, you should. . .

Before After
Study Study

1. understand that the planning and controlling functions at Sperry Van Ness are clearly separate.
Y, N, NI
2. recognize that controlling is primarily an organization-level concept and is not particularly relevant for lower-level employees.
Y, N, NI
3. realize that the first step in the controlling process is measuring performance.
Y, N, NI
4. understand that Sperry Van Ness must develop its own standards, which may or may not be the same standards as its competitors'.
Y, N, NI
5. emphasize to Sperry Van Ness's employees that problems and symptoms are often the same thing.
Y, N, NI
6. understand that managerial power is a function of an individual's position and personal relationships with others.
Y, N, NI
7. understand that conveying a high level of expertise will increase the level of power that Sperry Van Ness employees will associate with managers who demonstrate such expertise.
Y, N, NI

8. understand that information has value that depends on the appropriateness, quality, timeliness, and quantity of the information.
Y, N, NI
9. use the terms *data* and *information* interchangeably, because they are virtually the same.
Y, N, NI
10. update information systems as needed to ensure that Sperry Van Ness's managers have access to the information they need to make quality decisions.
Y, N, NI
11. recognize that the first step needed to operate an IS effectively entails summarizing data.
Y, N, NI
12. distribute all important information to all of Sperry Van Ness's managers, because these managers all need access to the same information.
Y, N, NI
13. not worry about an information system's ease of use; evidence suggests that users will learn to use even difficult systems if such use is necessary for their job performance.
Y, N, NI
14. recognize that the world is filled with IS professionals, and as such, traditional human resource management techniques are unnecessary for Sperry Van Ness's IS employees.
Y, N, NI
15. understand that information security continues to represent an important factor for companies such as Sperry Van Ness to consider as they maintain their information systems.
Y, N, NI

Exploring Your Management Skill: Part 2

As you recall, you completed Exploring Your Management Skill before you started to study this chapter. Your responses gave you an idea of how much you initially knew about controlling and helped you focus on important points as you studied the chapter. Answer the Exploring Your Management Skill questions again now and compare your score to the first time you took it. This comparison will give you an idea of how much you have learned from studying this chapter and pinpoint

areas for further clarification before you start studying the next chapter. Record your answers within the text or go to MyManagementLab.com. Recording your answers in MyManagementLab will allow you to get immediate results and see how your score compares to your classmates. If you answer the questions in the book, the answers are located in the Exploring Your Management Skill section at the end of the book.

Your Management Skills Portfolio

Your Management Learning Portfolio is a collection of activities especially designed to demonstrate your management knowledge and skill. By completing these online at MyManagement.com, you will be able to print, complete with cover sheet, as many activities as you choose. Be sure to save your work. Taking your printed portfolio to an employment interview could be helpful in obtaining a job.

*The portfolio activity for this chapter is Controlling at Merrill Lynch. Study the information and complete the exercises that follow.*⁵³

Merrill Lynch provides a number of financial services to corporations and individuals. One of Merrill Lynch's primary activities involves providing advisory services to individual investors. To provide these services, financial planners work with clients

and provide advice regarding potential investment decisions. In exchange for this advice, the financial planners earn money for the company based on commissions and other fees. Financial planners at Merrill Lynch work with different numbers of clients, and these clients vary dramatically in terms of their total assets.

John Thain, Merrill Lynch's CEO, has contacted you to help him think of ways to improve the performance of the many financial planners working for the company. His specific task for you involves making the controlling process relevant to the financial planners. In the following sections, answer the various questions as they pertain to Merrill Lynch. Visiting the company's Web site (www.ml.com) might help you as you think about this process.

1. The first step in the controlling process involves measuring performance. If you were John Thain, how would you measure the performance of the financial planners? Be specific.

2. The second step in the controlling process entails comparing measured performance to standards.

- a. What types of standards would you develop to help assess performance of these financial planners?

- b. What information sources would you use for developing these standards?

3. The final step in the controlling process involves taking corrective actions if necessary.

- a. What types of corrective actions would you take to help control the performance of financial planners?

- b. How would you determine whether corrective actions were necessary?

Experiential Exercises

1 Working with Information

Directions. Read the following scenario and then perform the listed activities. Your instructor may want you to perform the activities as an individual or within groups. Follow all of your instructor's directions carefully.

Perhaps the most critical aspect of an information system is determining what information a given organization needs to operate effectively. When an organization understands what information is needed, leaders can design an information system that will allow them to collect this information efficiently. In this exercise, your group should choose a local restaurant and then assume you are the top management team for this restaurant. With this restaurant in mind, determine the primary pieces of information you need to ensure the restaurant operates effectively. If possible, group these different pieces of information into logical categories.

2 You and Your Career

In the previous sections we provided an overview of the controlling process, which involves measuring performance, comparing performance to standards, and taking corrective action if necessary. During your career in both management and nonmanagement roles, you will deal with performance standards. How have standards played a role in your life so far? Examples might include your academic career, your current or previous employer, or even extracurricular activities such as participation in sports. Until now, who has defined the standards that apply to you, and what role have you played in this process? How might your familiarity with standards influence your career in the future?

Videonet Exercise

Controlling and Technology: Platinum Autobody

Video Highlights

Platinum Autobody is a small, family-owned business focusing on custom body and mechanical jobs. The co-owners discuss how the four functions of management apply to their business: planning, organizing, leading, and controlling. In the video, the co-owners define their managerial roles and division of responsibilities.

Discussion Questions

1. Saboor and Kawa Hatef divide responsibilities for running Platinum Autobody. What are the advantages of such specialization?
2. How is Platinum Autobody using technology to improve its business?

3. Why would Platinum Autobody compare its prices to those charged by competitors?

Internet Activity

Browse the Platinum Autobody site at platinumab.com. Click on the ASE Certified link and read about the company's facilities and personnel. Now that you know a little more about the company, discuss the four functions, such as how planning applies to expanding the business, how organization benefits from technology, how leading provides motivation for employees, and how controlling applies to all operations of the shop. How do the ASE and I-Car certifications contribute to the controlling aspect of Platinum Autobody?

CASES

1 SPERRY VAN NESS: HARNESSING TECHNOLOGY FOR BUSINESS SUCCESS

“Sperry Van Ness: Harnessing Technology for Business Success” (p. 501) and its related Challenge Case Summary sections were written to help you better understand the management concepts contained in this chapter. Answer the following discussion questions about this Challenge Case to further enrich your understanding of chapter content:

1. List three decisions an improved IS could help Sperry Van Ness president and CEO Kevin Maggiasco make. For each decision, describe the data that must be in the database to provide such help.
2. The main steps of the controlling process are measuring performance, comparing performance to standards, and taking corrective action. Discuss the possible role of an IS at Sperry Van Ness in each of these steps.
3. In addition to commercial real estate, some of Sperry Van Ness’s chief competitors also handle residential real estate. Because of this additional area of operation, do you think Sperry Van Ness’s IS should differ from that of its competitors? Why?

2 TECHNOLOGY HELPS NIKE RACE TOWARD HIGHER PERFORMANCE

Read the case and answer the questions that follow. Studying this case will help you better see how your understanding of information technology can be applied in a company such as Nike.

Nike takes a step forward in its race for sales and profits every time it sells a pair of sneakers. The Oregon-based company has a commanding lead over Adidas and other rivals because of its 32 percent share of the global athletic shoe market. Still, CEO Philip Knight and his executive team know they must sell a lot of shoes and sports apparel to expand beyond the current \$12 billion in yearly sales and hike the gross profit margin above 43 percent (the highest margin in company history). They have to estimate how well each product will sell, allow enough time to manufacture the right quantities, get the products to the right place at the right time at the right cost, and reach out to the right customers. The unpredictability of fashion trends makes their decisions even more challenging.

So how does Nike do it? With technology, a healthy dose of human expertise, and patience. The company was using 27 different information systems to handle sales forecasting, factory orders, and deliveries to retailers worldwide when management began planning for a single, integrated IS. One goal was to slash, from nine months to six months, the time needed to get shoes and other items from the design stage to store shelves. Another goal was closer coordination with the Asian factories that manufacture Nike’s shoes as a way to minimize inventory. This coordination would help the company avoid some of the financial risks of catering to fashion-conscious customers whose

tastes can change overnight. Finally, management wanted a single, centralized system for forecasting and ordering that managers and employees could learn to use efficiently and effectively.

Months of preparation went into the project, as company managers worked with specialists to customize software to Nike’s unique situation and then implemented the system before a new corporate-wide IT project took effect. At first, the forecasting/ordering system struggled to handle the more than 10 million stock numbers needed to track all product variations; it also operated more slowly than expected when tied to the existing IT network. Then the system issued factory orders for too many of some models and too few of others, causing Nike to lose an estimated \$100 million worth of sales.

The project managers created workarounds to make data available for planning purposes while they analyzed what had happened, revamped the system design, and got ready to implement it more gradually. “Once we got into this, we quickly realized that what we originally thought was going to be a two-to-three-year effort would be more like five to seven [years],” observes Roland Wolfram, Nike’s vice president of global operations and technology. In fact, the new system was phased in, area by area, over the next two years, and users received intensive training in advance. As a double check, Nike managers carefully scrutinize system output “to make sure it makes sense,” says Wolfram; they also ask retailers for input when forecasting demand for new items. Now, six years after the original project began, the system is working so well that Nike managers have set a new goal of reducing the lead time for orders even further.

The Internet plays a major role in Nike’s plans for communicating with customers to build brand loyalty and stimulate demand. For example, Nike keeps its trademark “swoosh” symbol in front of soccer fans by inviting them to log onto the company’s Web site and participate in multiplayer games such as Football Olé. It has teamed up with MSN Europe to create an Instant Messenger version of the same game that can be played by people across the continent. In addition, Nike has developed a series of Web sites tailored to the interests of specific customer groups, such as www.nikegoddess.com for women, www.nikerunning.com for runners, and www.nikebasketball.com for basketball fans. Watch for more technological innovation as Nike continues its never-ending race for higher sales and higher profits.

QUESTIONS

1. Why are information quality and timeliness particularly important to Nike’s success?
2. What security issues do you think Nike management should take into account when planning, designing, and implementing a system for sales forecasting and factory orders? Why?
3. What else should Nike do to use the World Wide Web for communicating with customers?

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Exploring Your Management Skill Answers

Chapter 1

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Chapter 14

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Chapter 15

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Chapter 16

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Chapter 17

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Chapter 18

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Chapter 19

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Chapter 20

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Chapter 21

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Chapter 22

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